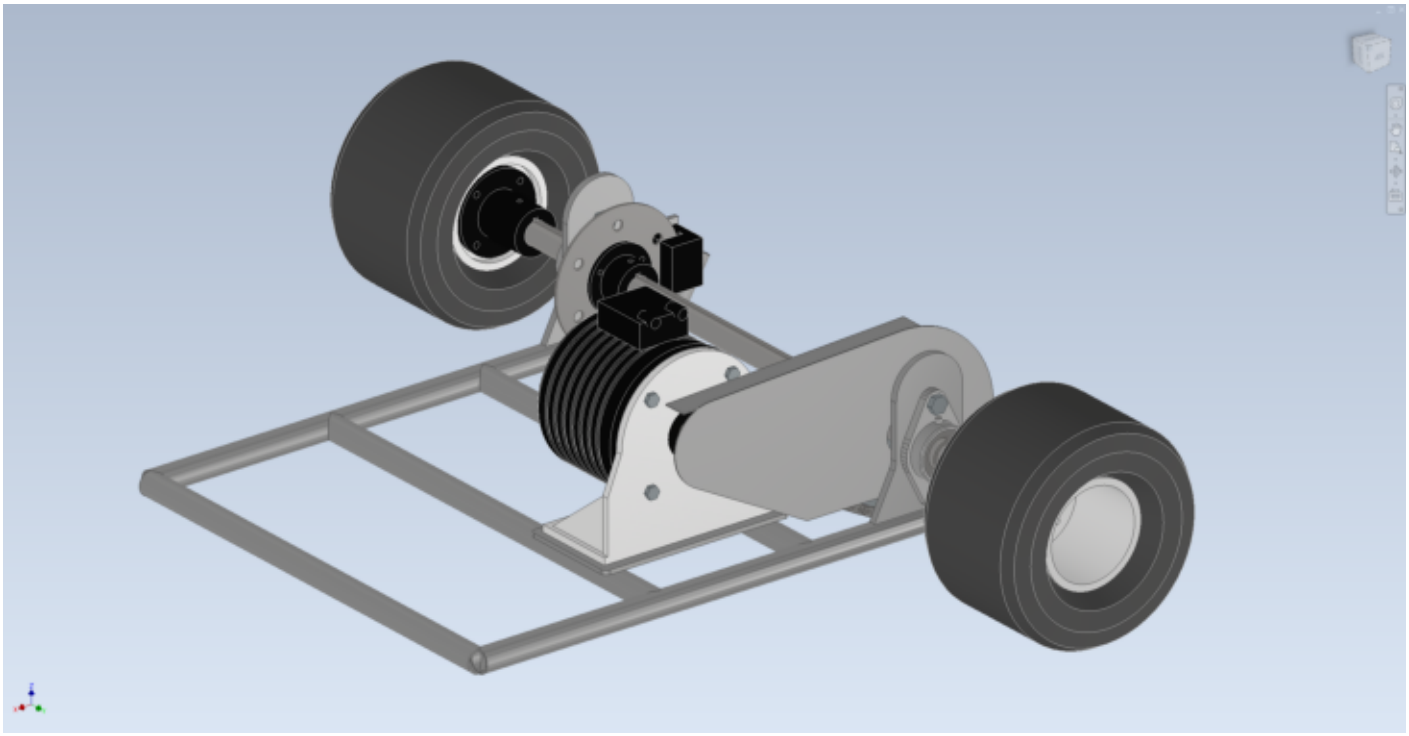
# GoKartDrivetrain - calculation report

Electric go-kart rear drivetrain for 25 mph. Jay Patel, Stevens Institute of Technology (ME).

Designed and simulated only: every number is calculated or simulated from the CAD model; nothing has been built or tested.



Kart 185 kg with driver, 60 % rear, CG 280 mm, 11x7.10-5 rear tyres (r roll 137 mm)

Motor Motenergy ME1114 PMSM (0.12 Nm/A, 65 Nm peak, 5000 rpm), 350 A controller -> 41.7 Nm; 16S LiFePO4 40 Ah

Reduction ANSI 40 chain 13/50 = 3.846:1, 78 links, centres 285.4 mm; 25 mph = 2996 rpm motor

Torque at 25 mph: needs 3.01 Nm, available 33.1 Nm (margin 11.0x nominal, 10.2x low battery)

Simulated 0-25 mph 2.14 s (nominal battery; 2.18 s low), peak 0.56 g, free top speed 32.5 mph (governed at 25)

Chain tension 1572 N at full torque -> SF 8.6 on min. ultimate; speed 8.2 m/s

Axle 1-1/4 4140 PH: static SF 4.85 (nominal) / 2.86 (with keyway Kt), fatigue SF 2.14 (DE-Goodman)

Bearings UC206-20: L10 61,556 h (left, worst) at 18 mph average, static SF 3.19

Inventor: 61 occurrences, 180 constraints, 0 DOF, 0 interference; drivetrain 30.5 kg + wheels 5.4 kg

FEA Motor-mount bracket (6061-T6): worst C1 launch: 39 MPa peak / 21 MPa p99.9, SF 6.99 / 13.12

FEA Bearing hanger (4130): worst C4 kerb 2g+1g L: 123 MPa peak / 90 MPa p99.9, SF 3.74 / 5.13

FEA Rear axle (4140 PH): worst C4 kerb 2g+1g: 139 MPa peak / 133 MPa p99.9, SF 4.70 / 4.94

# 1 Requirements, concept and gearing

Brief: an electric go-kart drivetrain in Autodesk Inventor capable of 25 mph: electric motor, motor mounting bracket, reduction drive, rear axle, bearings, sprockets, chain and chassis mounting interfaces, with motor-torque, gear-reduction, axle-loading and acceleration calculations, FEA of the motor mount and axle supports, commercially available parts, a complete assembly, manufacturing drawings and a BOM.

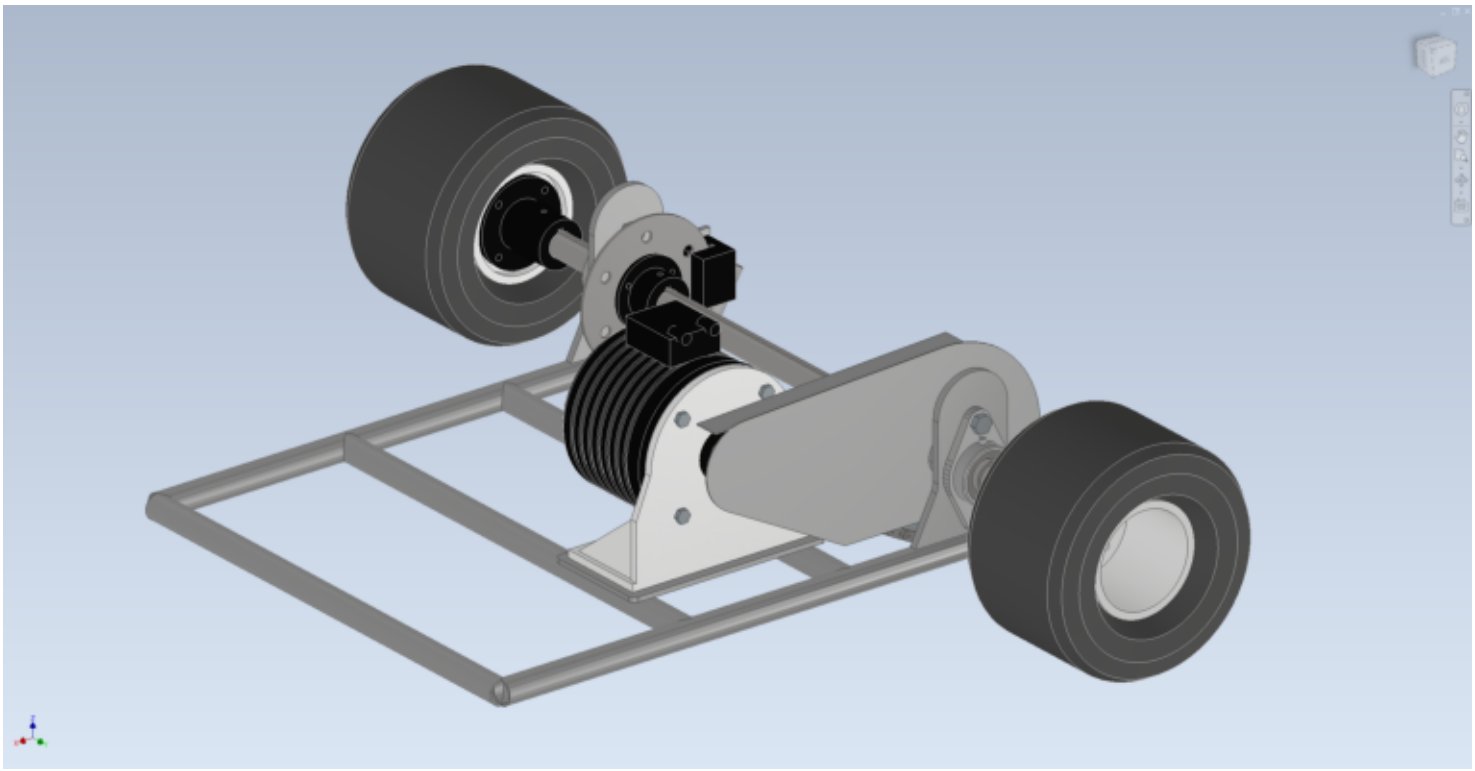
Respec (2026-10-29, after a buildability audit against real catalogue ratings): the first design used ANSI 35 chain (13T/60T), which ran at 99-126 % of its working allowable at launch, had no brake and drew a 5 in wide rim that an 11x7.10-5 tyre does not fit. This report is the respecced design: ANSI 40 chain on stock Martin sprockets, a hydraulic rear disc brake, a chain guard, 5 x 8 in rims on JOES 25560-pattern hubs and 16 mm UCFL206 bolt holes.

Concept: a live-axle sprint/rental-class kart (11 in rear wheels) driven from the left by a single-stage ANSI 40 chain. The motor is a Motenergy ME1114 axial-gap PMSM (a common electric-kart motor) bolted by its face to a machined 6061-T6 bracket. The bracket sits in four 24 mm slots on an A36 base plate welded across two chassis cross tubes, so sliding it forward tensions the chain (assembly parameter ChainAdjust). The 1-1/4 in 4140 axle runs in two UC206-20 insert bearings in UCFL206 flange housings, bolted to 3/8 in 4130 hangers saddled onto the main rails. The Martin 40BS50 sprocket (1-1/4 bore) is keyed straight to the axle. Its plane and the 13T motor sprocket are driven by the same parameter (ChainLine), so the two stay coplanar by construction.

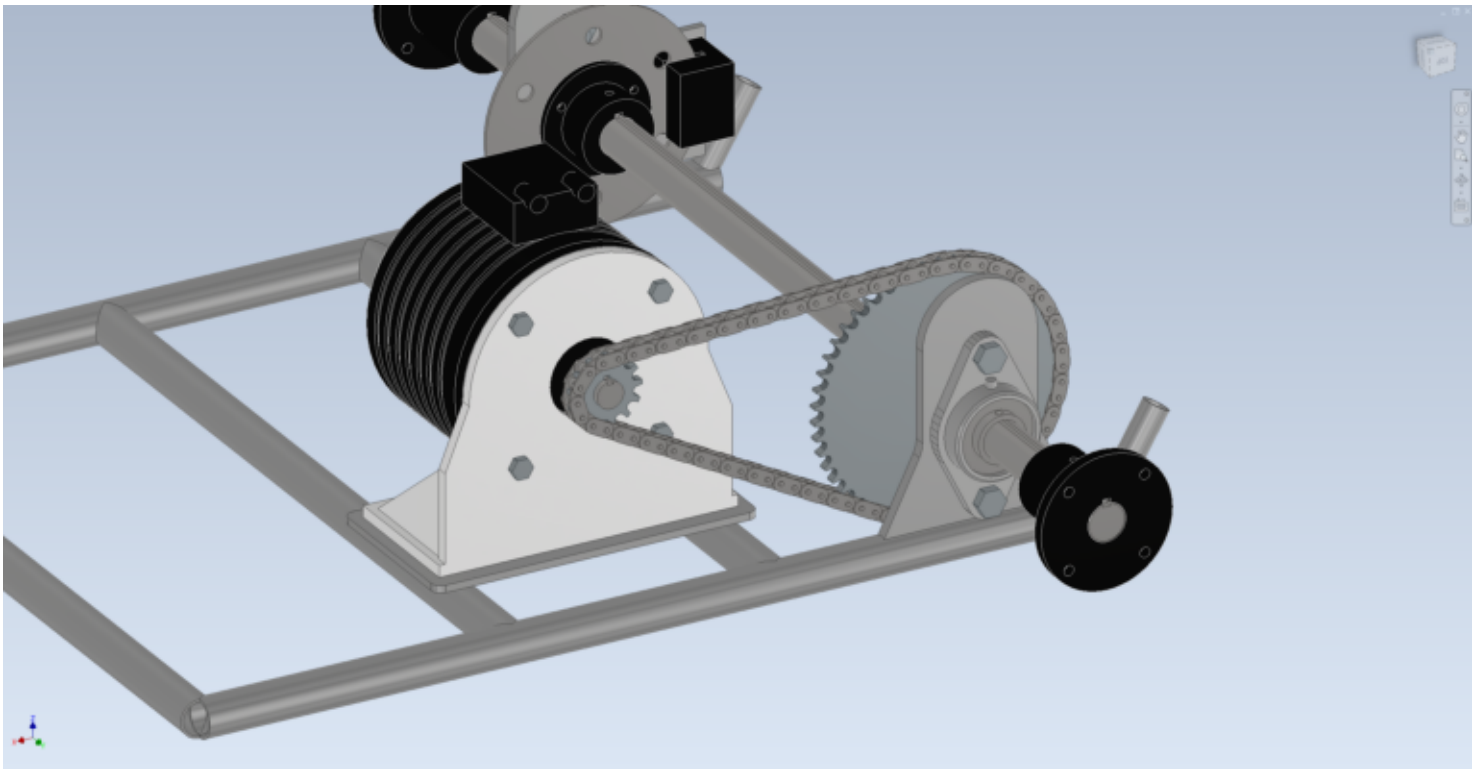
Gearing logic: top speed is set by the gearing, not by power. 25 mph at the 137 mm rolling radius is 779 rpm at the axle; the ME1114's back-EMF (Ke = Kt = 0.12 V s/rad) limits it to ~3664 rpm at a nearly flat pack (49.6 V). So the ratio was chosen to reach 25 mph even at low charge: 13/50 = 3.846 puts 25 mph at 2996 rpm (60 % of the 5000 rpm limit), and the controller's speed limit holds 25 mph on a full pack (free speed 33.5 mph). Both sprockets are catalogue parts: Martin 40BS13 with a 7/8 in bore on the motor, 40BS50 with a 1-1/4 in bore on the axle; the 50T leaves 32 mm of ground clearance.

Chain geometry: the 78-link loop was closed exactly (chordal pitch-point march, closure error < 1e-12 mm), giving centres 285.399 mm. Both sprockets' tooth gaps are phased to the chain pitch points, so the modelled chain seats in the teeth (30 pitch points wrap the 50T and 6 wrap the 13T, wrap angle 150 deg).

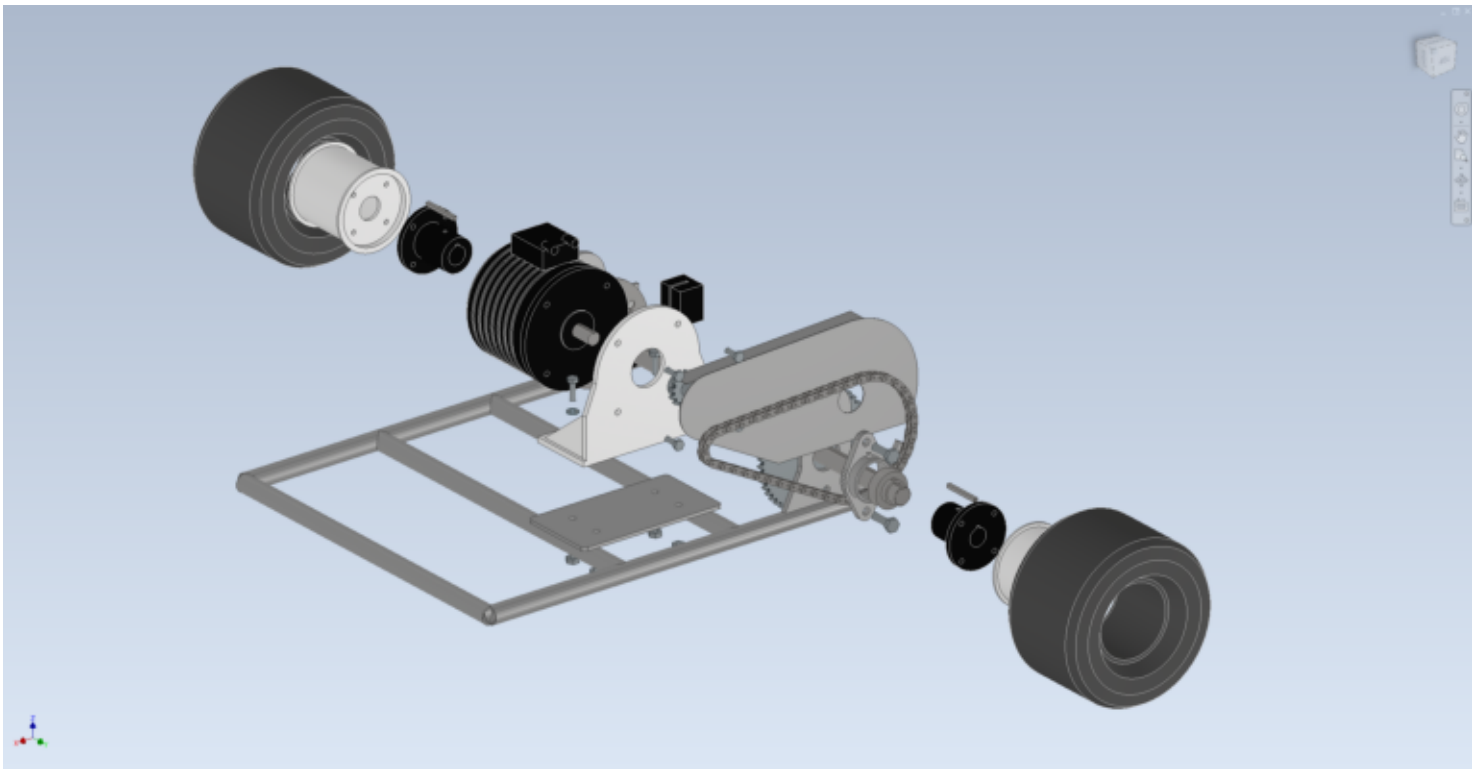
## Isometric (Inventor, Realistic style)



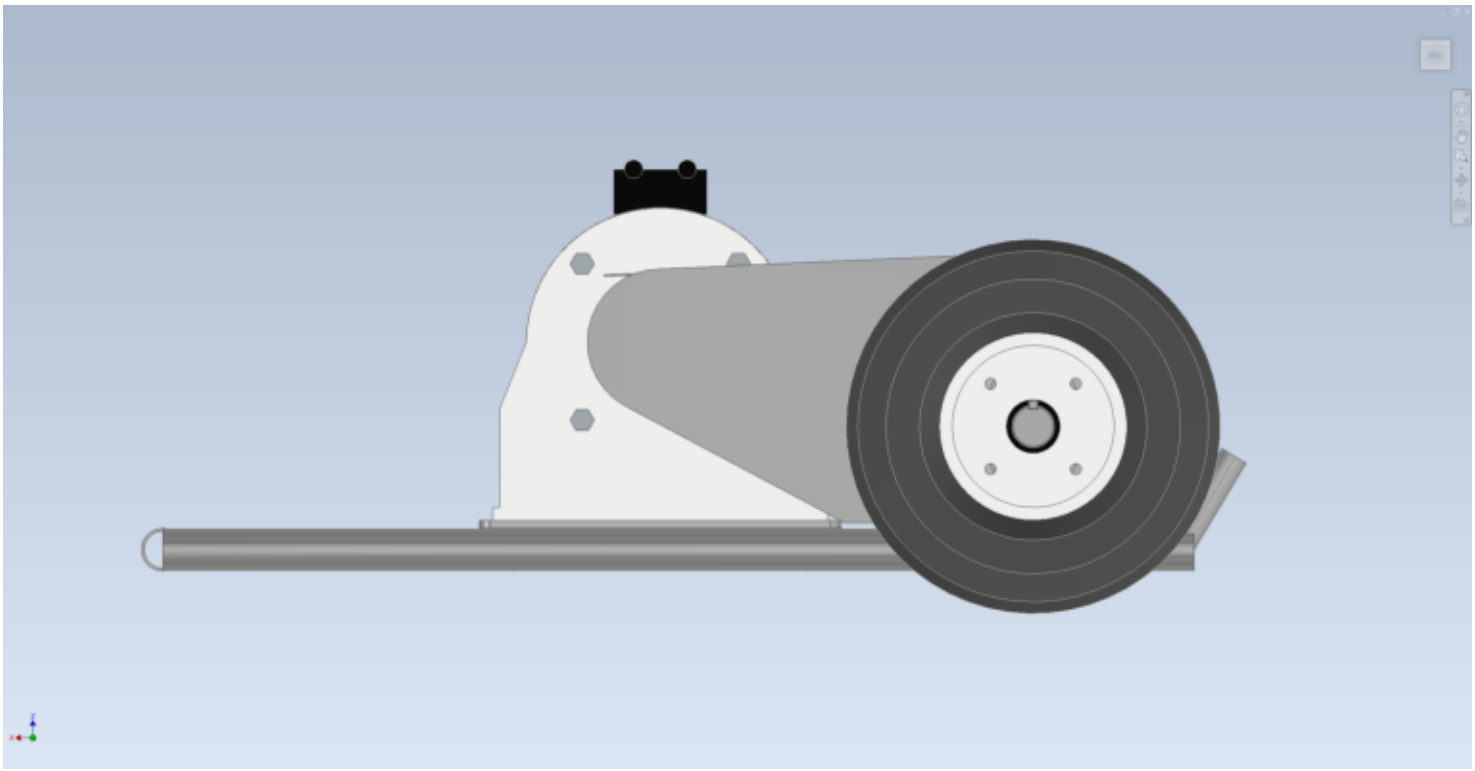
**Drive close-up: 13T/50T #40 chain, slotted motor bracket, hanger**



## Exploded view (GoKartDrivetrain_Exploded.iam)



## Left side (chain side)



# 2 Motor torque, road load, acceleration (simulated)

Road load at 25 mph: rolling 32.7 N (Crr 0.018) + aero 40.5 N (CdA 0.54 m2) = 73.1 N, 817 W at the wheels; referred to the motor 3.01 Nm. Cruise battery power 1059 W, so range at a steady 25 mph is ~62 km at 80 % depth of discharge (simulated).

Motor model: shaft torque = Kt I - 0.3 Nm drag. Phase current is limited by the controller (350 A), the battery (250 A battery-side) and the voltage (Ke w + R I <= 0.95 V_batt, with a 30 mOhm pack sag). The ME1114 peak stall torque (65 Nm) is above the 350 A controller limit (42 Nm), so the controller sets the launch torque.

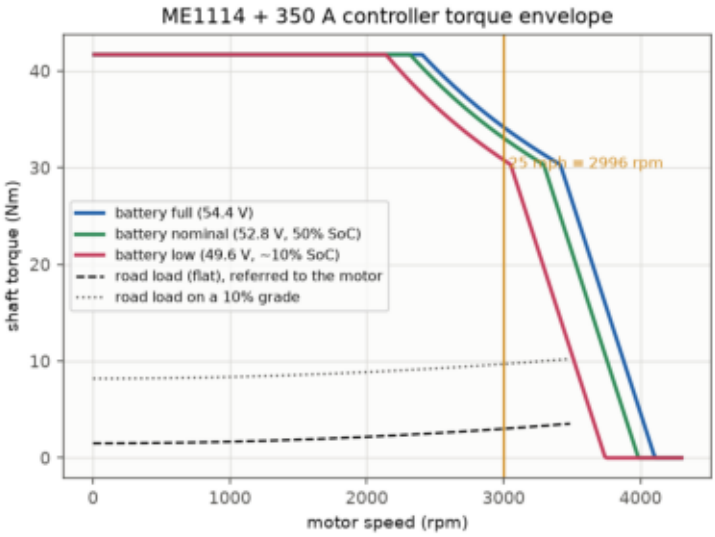
Launch simulation: explicit Euler at 1 ms. Effective mass 199.7 kg includes the wheels, axle and motor rotor (rotor 0.012 kg m2 assumed) reflected through the ratio. The traction limit is mu (1.1) times the rear load including longitudinal transfer; it never binds, because full motor torque gives about 1.2 kN at the wheels against ~1.6 kN of grip. Results (s, m, g, kW):

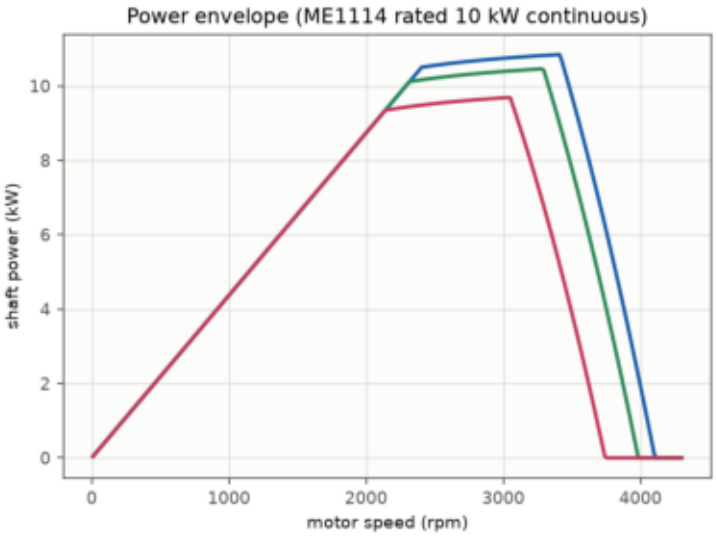
| battery | T@25mph | margin | v free | 0-10 | 0-20 | 0-25 | d 0-25 | a max | P batt |
|---|---|---|---|---|---|---|---|---|---|
| full | 34.2 | 11.4x | 33.5 | 0.82 | 1.65 | 2.12 | 12.2 | 0.56 | 11.7 |
| nominal | 33.1 | 11.0x | 32.5 | 0.82 | 1.65 | 2.14 | 12.3 | 0.56 | 11.3 |
| low | 30.8 | 10.2x | 30.6 | 0.82 | 1.66 | 2.18 | 12.8 | 0.56 | 10.5 |

Gradeability (nominal battery, steady speed the drive can hold, governor ignored): 5% 31.9 mph, 10% 31.3 mph, 15% 30.7 mph, 20% 30.1 mph, 30% 29.0 mph. Start-from-rest limit 76 % (traction-limited, theoretical).

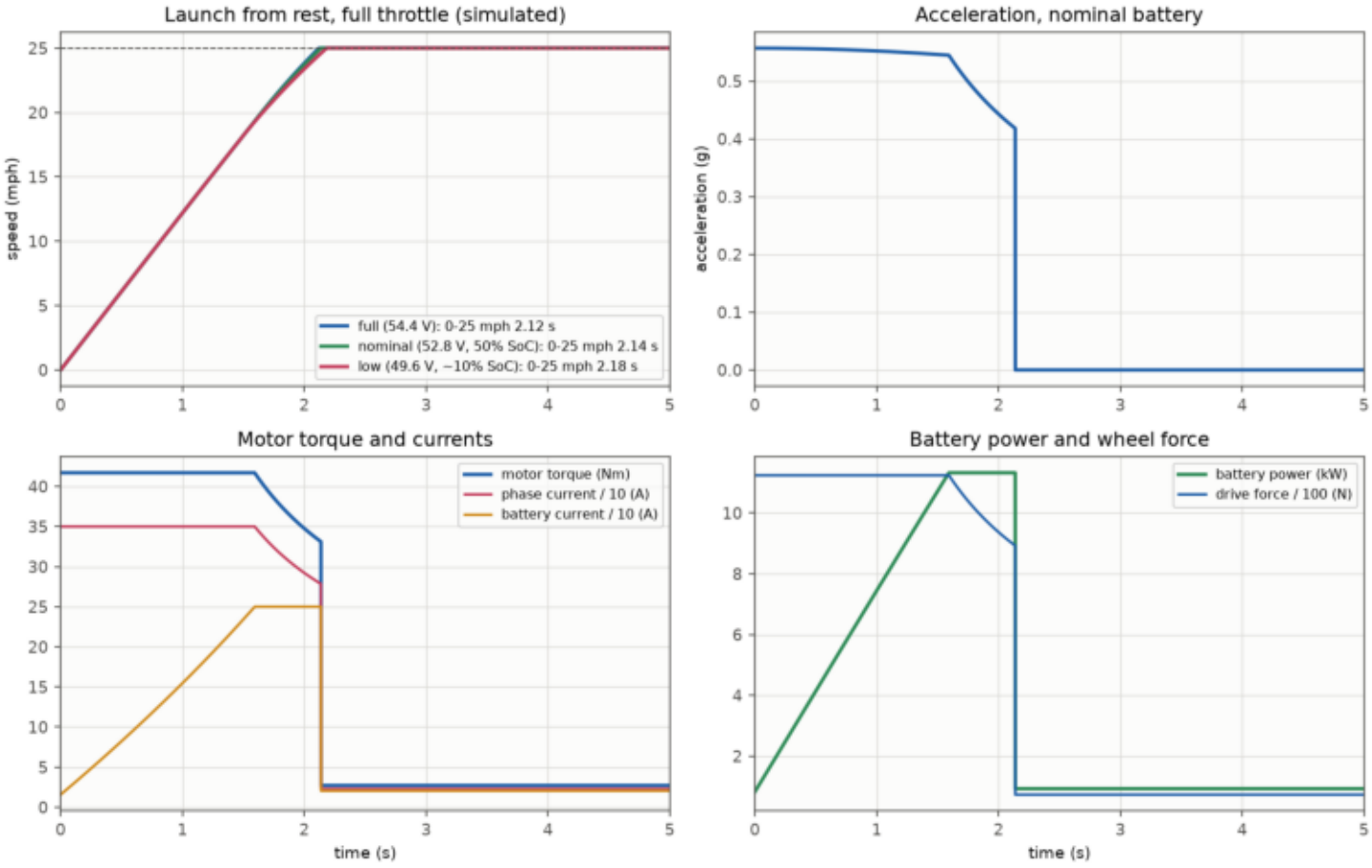
Torque margin: launch torque 41.7 Nm is 14x the 25 mph requirement; at 25 mph the voltage-limited torque available is 11.0x (nominal) and 10.2x (low pack) the road load, so 25 mph is reached at any state of charge. Cruise uses cruise 1.06 kW vs 10 kW continuous.

# Motor + controller torque / power envelope vs road load

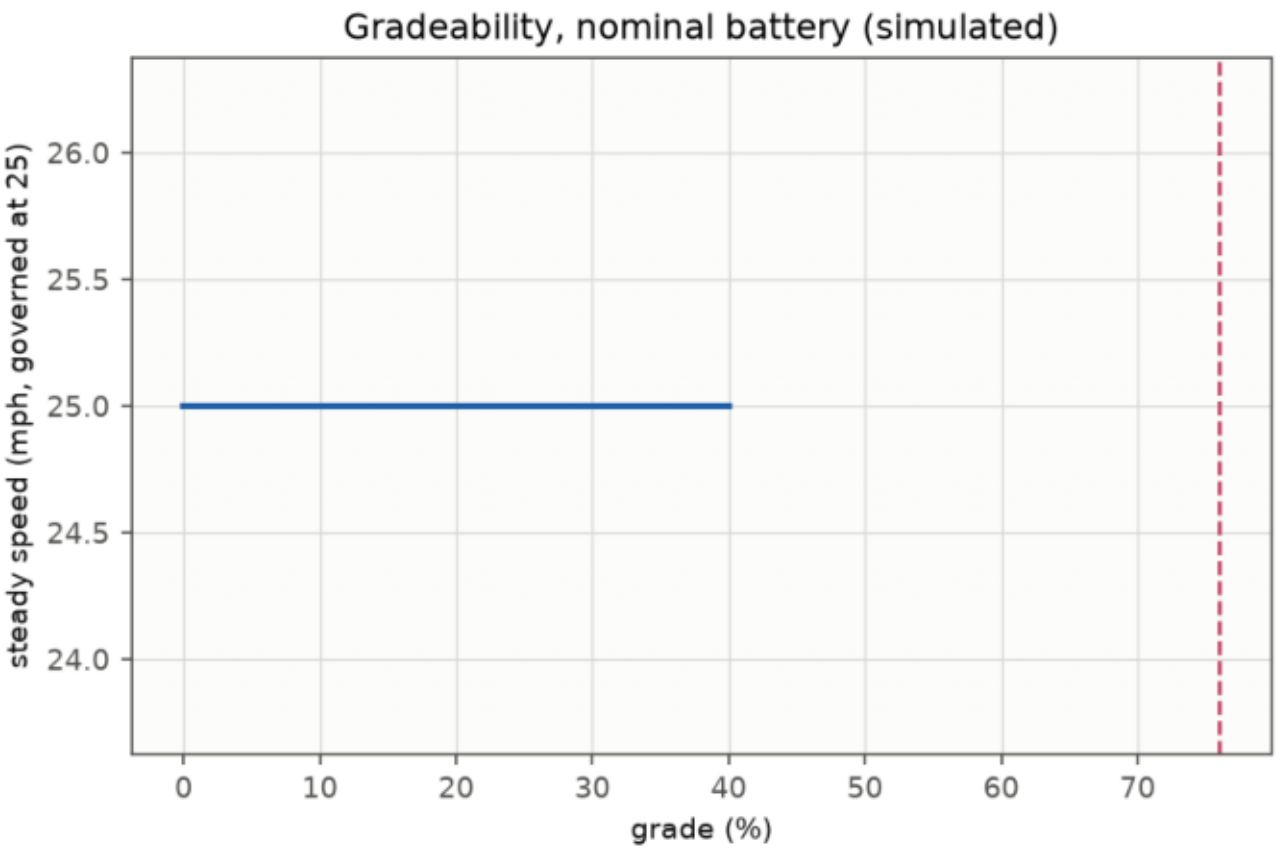


# Launch simulation



## Gradeability



# 3 Chain, sprockets, keys

Chain: ANSI 40 roller chain, single strand, pitch 12.7 mm, 78 links (0.991 m), min. ultimate tensile 13.90 kN (ANSI B29.1). At 25 mph it runs at 8.24 m/s (1623 fpm), centrifugal tension 41 N.

Tight-span tension = T_motor / r_pitch(13T): 1572 N at the 42 Nm launch (SF 8.62 on min. ultimate); 2238 N if the tyres could transmit 0.75 g with mu 1.1 (SF 6.10); 114 N at 25 mph cruise (SF 90).

Power rating (Shigley eq. 17-32/33, 13T, roller/bushing-limited at 2996 rpm): 2.08 kW for 15,000 h. Cruise power x Ks 1.3 = 1.23 kW (rating SF 1.69). Full-power bursts (10.5 kW shaft) are 6.5x the 15,000 h rating: like every kart, chain wear is a maintenance item (inspect and replace at 3 % elongation). ANSI type B (bath / disc) at > 1300 fpm; on a kart: drip/spray lube each session.

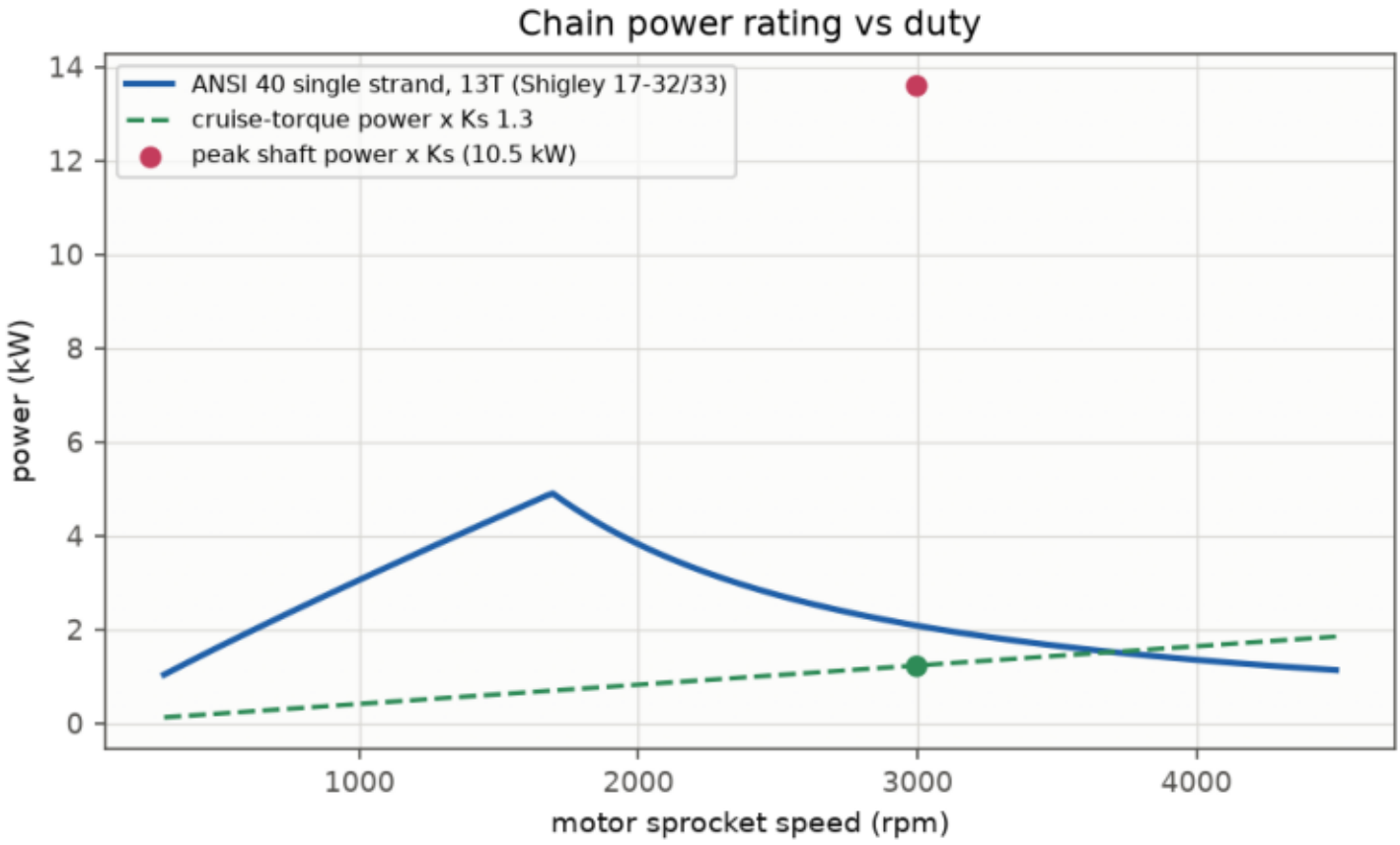
Tensioning: the 4 slots give +-12 mm, i.e. centres 272.2 to 295.6 mm (measured in Inventor), enough to take up ~2.5 % chain stretch. Slack-span sag vs centre reduction: 0 mm: 0 mm, 2 mm: 20 mm, 4 mm: 28 mm, 6 mm: 35 mm (set about 10-12 mm sag at the slack span).

Sprocket teeth (whole tension on one tooth, conservative): root shear 47 MPa -> SF 6.54 (13T) / 3.83 (50T, Sy 310 MPa assumed). Working load: launch 1572 N = 44 % and traction limit 2238 N = 62 % of the 3.60 kN Tsubaki RS40 maximum allowable load (the first #35 design reached 99-126 %). The axle sprocket is keyed to the axle, so there are no sprocket bolts.

Keys (1018 CD, Sy 370 MPa; Ssy = 0.577 Sy; crushing on half the key height):

| key | T Nm | shear | crush | SF shear | SF crush |
|---|---|---|---|---|---|
| axle sprocket key 1/4 x 1/4 x 25 (40 | 157 | 62 | 125 | 3.42 | 2.96 |
| wheel hub key 1/4 x 1/4 x 70 (whole | 157 | 22 | 45 | 9.57 | 8.30 |
| motor sprocket key 3/16 x 3/16 x 19 | 42 | 41 | 83 | 5.15 | 4.46 |

## Chain power rating vs duty



# 4 Axle loading, bearings, mounts, brake

Load cases (quasi-static, at the rear tyres): C1 launch: full motor torque 350 A (42 Nm), 0.75 g launch with load transfer; C2 bump 2.5g: 2.5 g vertical on both rear wheels at 25 mph cruise torque; C3 corner 1.5g: 1.5 g right turn, inner (right) rear lifted, outer (left, sprocket side) wheel carries the whole rear load and side force, 50 % torque on exit; C3R corner 1.5g: mirror of C3: left turn, outer = right wheel; C4 kerb 2g+1g: 2 x rear-axle load on the left wheel (kerb strike), 1.0 g side load, full torque.

Axle statics: wheel contact-patch forces, the chain pull on the 50T (tight upper span, toward the motor) and two bearing reactions (radial at each insert, axial shared 50/50 by the set-screw-locked inserts). Bending and torque diagrams are integrated along the axle at 1 mm. Forces N, torque / moment N m, stress MPa:

| case | chain T | axle T | Fz L | Fz R | Fy L | Fy R | brg L Fr | brg R Fr | Mb max | vm | SF |
|---|---|---|---|---|---|---|---|---|---|---|---|
| C1 launch | 1572 | 157 | 728 | 728 | 0 | 0 | 2066 | 1050 | 213 | 71 | 9.18 |
| C2 bump 2.5g | 114 | 11 | 1361 | 1361 | 0 | 0 | 1364 | 1362 | 313 | 100 | 6.57 |
| C3 corner 1.5g | 786 | 79 | 1089 | 0 | -1633 | -0 | 1832 | 114 | 170 | 58 | 11.25 |
| C3R corner 1.5g | 786 | 79 | 0 | 1089 | 0 | 1633 | 495 | 1422 | 170 | 58 | 11.23 |
| C4 kerb 2g+1g | 1572 | 157 | 2178 | 544 | -1089 | -0 | 3537 | 264 | 410 | 135 | 4.85 |

Axle stress: section modulus of the 31.75 mm bar; the full-length keyway is taken as sled-runner type (Kt 1.7, Kts 1.6, Shigley Table 7-1). Static SF on 4140 PH (Sy 655 MPa): 4.85 nominal, 2.86 with Kt (worst C4 kerb, at the bearing). Fatigue (rotating bending, steady torque, DE-Goodman, Se = 256 MPa from Marin ka 0.72, kb 0.86, 99 % reliability): SF 2.14 (C1 launch at the bearing), 2.65 for the every-corner 1.5 g case.

Axle deflection (simply supported at the bearings): C2 bump 2.5g: wheel 2.16 / 2.16 mm, slope 0.77 deg; C4 kerb 2g+1g: wheel 1.95 / 1.31 mm, slope 0.67 deg; C3 corner 1.5g: wheel 0.05 / 0.06 mm, slope 0.03 deg. Kart axles flex by design; UC inserts accept +-2 deg misalignment.

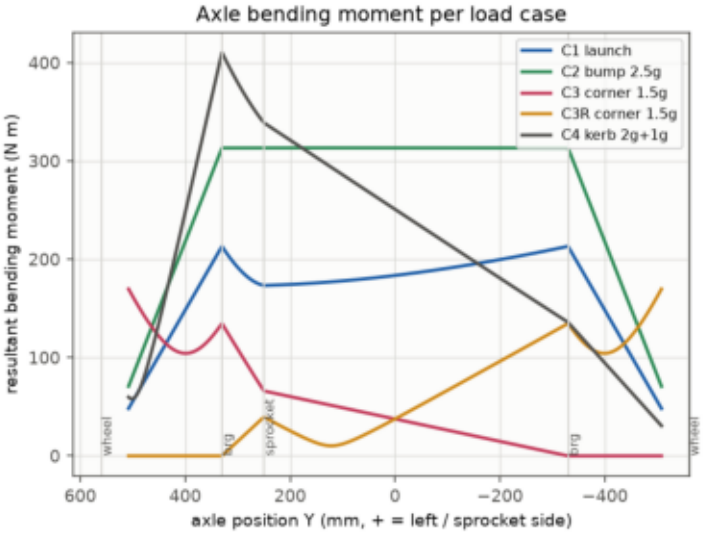
Bearings UC206-20 insert bearing (1-1/4 in bore) in UCFL206 2-bolt flange housing (C 19.5 kN, C0 11.3 kN). Duty cycle cruise 55%, C3 corner 1.5g 15%, C3R corner 1.5g 15%, C1 launch 12%, C2 bump 2.5g 3% at an 18 mph average (561 rpm): equivalent load 1530 N left / 1254 N right -> L10 61,556 h / 111,745 h; static SF C0/P0 3.19 / 7.95 (worst C4 kerb).

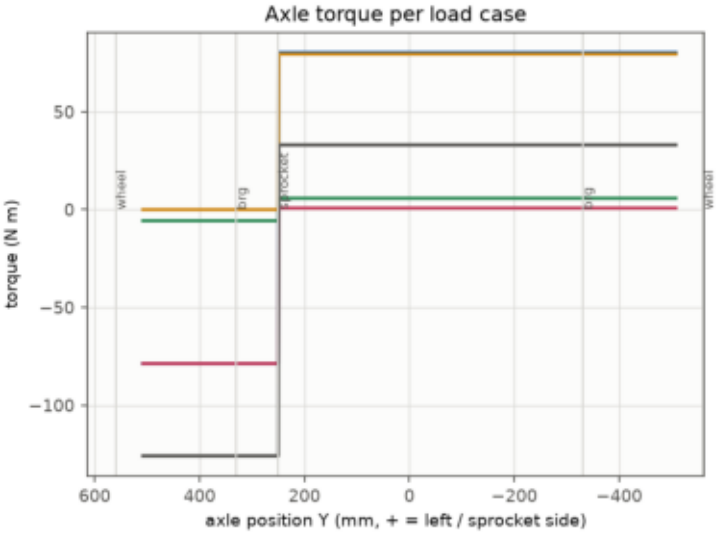
Motor mount: slot bolts 4 x 3/8-16 grade 8 at 75 % proof (31.0 kN each), friction mu 0.2 -> slip SF 14.9 against the 1667 N in-plane load, so the bracket cannot creep in its slots. Hanger welds (2 x 5 mm TIG fillet welds (ER70S-2) along the tube, 150 mm each side): 22 MPa, SF 6.5 vs the AISC 0.3 x 480 MPa allowable. UCFL206 bolts: the housing has the catalogue 16 mm holes; 1/2-13 grade 8 bolts go through 14 mm holes in the hanger with F436 hardened washers under the nuts.

Brake (added in the respec): rear-axle-only braking at the grip limit decelerates 0.51 g and needs 127 N m at the axle. With pad mu 0.40, 30 mm piston, 5/8 in master cylinder, 5:1 pedal (vendor data not published), the pads need 2038 N of clamp at a 77.7 mm pad radius: 29 bar and 114 N on the pedal. A 445 N (100 lbf) panic push gives 492 N m (SF 3.9). One stop from 25 mph puts 12.5 kJ into the 0.55 kg disc (+49 C). Brake-hub key 1/4 x 25: 50 MPa shear, SF 4.2.

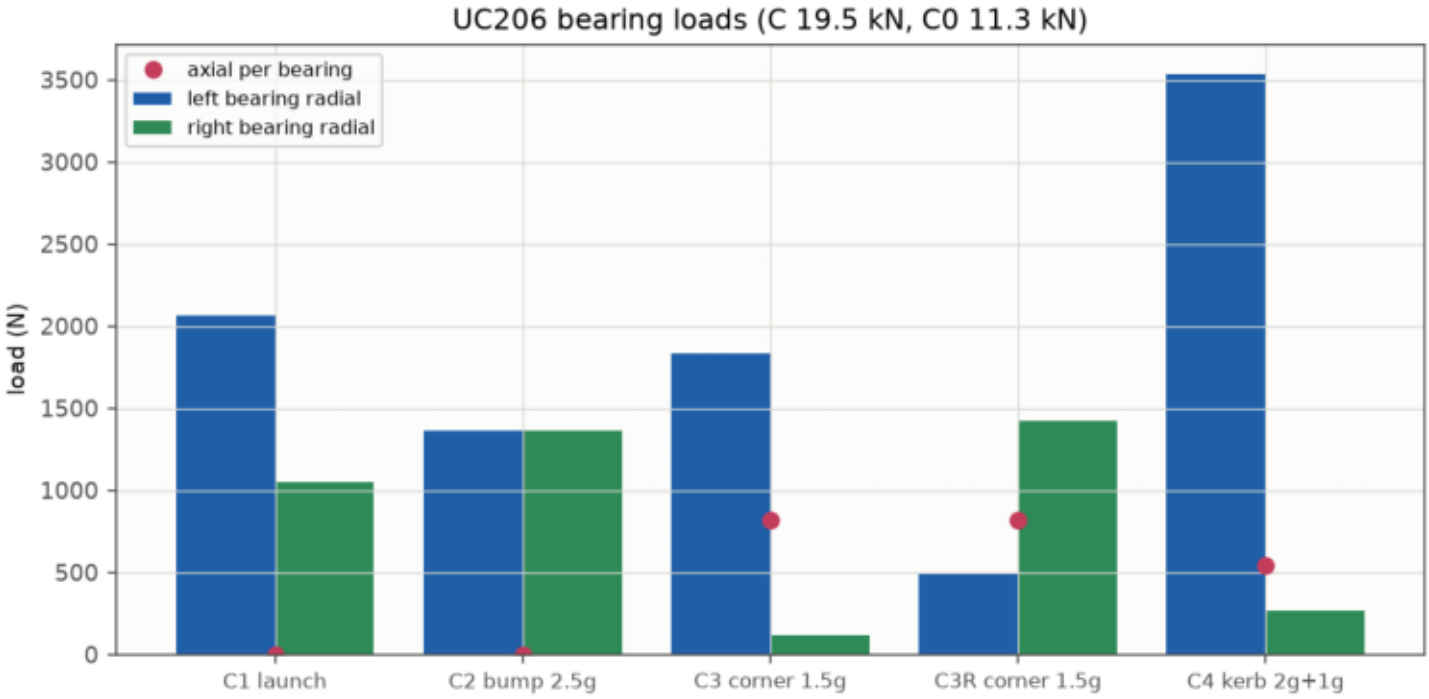
Caliper bracket (1/4 in A36 bracket, 2 x 5 mm TIG fillets (ER70S-2) along its two 105 mm edges on the right hanger): pad drag 1630 N acting 31.5 mm from the weld plane -> 5.6 MPa in the welds (SF 26); M8 bolt bearing in the plate 16 MPa (SF 23).

# Axle bending moment and torque per load case



## Bearing loads per case



# 5 Structural FEA (simulated)

Solver: gmsh (OpenCASCADE STEP import of the Inventor parts) + scikit-fem linear elasticity, quadratic P2 tetrahedra, PyPardiso direct solve with all load cases as one multi-RHS solve (same chain as F1Suspension / QuadDrone). An AMG-preconditioned CG fallback exists for low-RAM runs; the final results all used the direct solver.

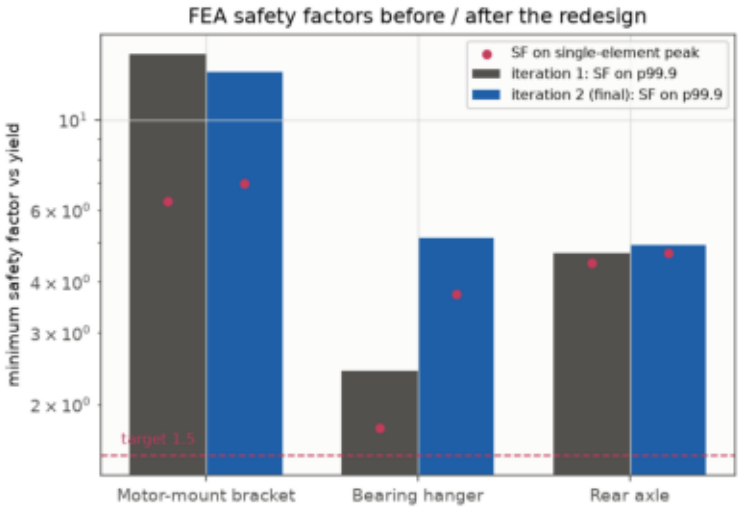
Motor bracket (6061-T6, Sy 276 MPa): the motor's in-plane load (chain pull + inertia) and its torque go through the four motor-bolt holes as cosine bearing pressure. The out-of-plane force and two bending moments act as a linear pressure on the motor-face contact ring (moments taken about the face, minus the bolt-plane offset). The four slot-bolt clamp areas under the flange are fixed. Bearing hanger (4130, Sy 460 MPa, as-welded HAZ allowable 360 MPa): the bearing force acts at the insert centre, 15 mm outboard of the plate face. Its in-plane part goes through the two 1/2 in bolt holes; the axial force and offset moments act as a linear pressure on the flange footprint. The saddle (weld to the rail) is fixed. Ten columns: the left-hanger loads of all five cases, and the right-hanger loads mirrored onto the same part, with 100 % of the axial load on the analysed bearing. Axle (4140 PH, Sy 655 MPa): wheel forces as force + bending couple + torque on the hub seats, chain load on the sprocket-hub seat, bearing reactions on the insert seats. The set is exactly self-equilibrated except the 1 % bearing-friction torque, so a 3-2-1 support carries <= 153 N. Torque on the keyway-interrupted seats is applied as force-free shear (the keys are not modelled).

Design iteration driven by the FEA (iteration 1 -> 2): the 1/4 in hanger reached 262 MPa peak / 190 MPa p99.9 (SF 1.75 / 2.42) at the re-entrant taper kink. It deflected 3.0 mm out of plane under the 2 g kerb + 1 g side case, too flexible for the bearing line. It became 3/8 in with R25 blends at the kinks: 129 / 93 MPa (SF 3.57 / 4.94), deflection 0.92 mm, +0.43 kg each. The 1/2 in motor bracket was far oversized (p99.9 SF 14.5) and became 3/8 in: 2.51 -> 1.96 kg, still p99.9 SF 10.8.

Peak vs design value: sharp CAD edges (slots, bores, weld toes) give single-element singularities. Both the peak and the 99.9th-percentile element stress are reported, and both meet the 1.5 target on yield at limit load.

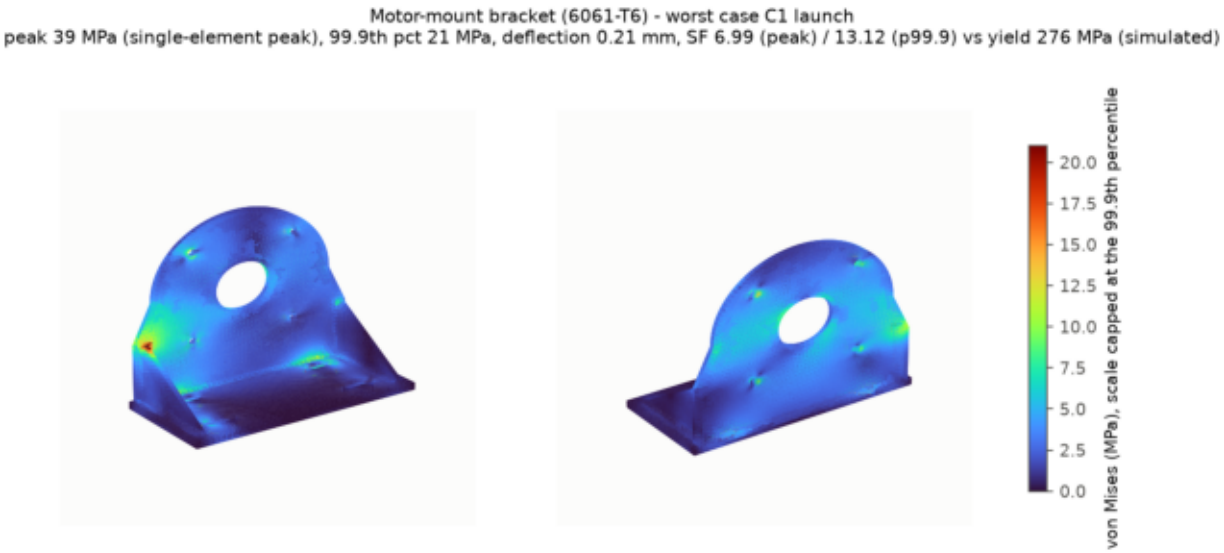
| part | case | DOF | peak | p99.9 | u | SF peak | SF p99.9 |
|---|---|---|---|---|---|---|---|
| MotorMountBrac | C1 launch | 275,304 | 39 | 21 | 0.21 | 6.99 | 13.12 |
| MotorMountBrac | C2 bump 2.5g | 275,304 | 26 | 12 | 0.37 | 10.54 | 23.98 |
| MotorMountBrac | C3 corner 1.5g | 275,304 | 18 | 11 | 0.06 | 15.20 | 26.00 |
| MotorMountBrac | C3R corner 1.5g | 275,304 | 36 | 16 | 0.35 | 7.64 | 16.77 |
| MotorMountBrac | C4 kerb 2g+1g | 275,304 | 35 | 21 | 0.24 | 7.98 | 13.45 |
| BearingHanger | C1 launch L | 263,118 | 32 | 27 | 0.14 | 14.17 | 17.24 |
| BearingHanger | C1 launch R | 263,118 | 20 | 16 | 0.13 | 23.45 | 28.80 |
| BearingHanger | C2 bump 2.5g L | 263,118 | 28 | 24 | 0.23 | 16.39 | 19.45 |
| BearingHanger | C2 bump 2.5g R | 263,118 | 28 | 24 | 0.23 | 16.40 | 19.26 |
| BearingHanger | C3 corner 1.5g L | 263,118 | 116 | 84 | 0.91 | 3.95 | 5.45 |
| BearingHanger | C3 corner 1.5g R | 263,118 | 85 | 63 | 0.73 | 5.41 | 7.35 |
| BearingHanger | C3R corner 1.5g L | 263,118 | 87 | 64 | 0.73 | 5.31 | 7.21 |
| BearingHanger | C3R corner 1.5g R | 263,118 | 112 | 83 | 0.91 | 4.12 | 5.57 |
| BearingHanger | C4 kerb 2g+1g L | 263,118 | 123 | 90 | 0.91 | 3.74 | 5.13 |
| BearingHanger | C4 kerb 2g+1g R | 263,118 | 54 | 39 | 0.45 | 8.55 | 11.95 |
| RearAxle | C1 launch | 411,261 | 74 | 71 | 2.31 | 8.88 | 9.17 |
| RearAxle | C2 bump 2.5g | 411,261 | 108 | 107 | 4.29 | 6.09 | 6.14 |
| RearAxle | C3 corner 1.5g | 411,261 | 49 | 46 | 0.56 | 13.41 | 14.18 |
| RearAxle | C3R corner 1.5g | 411,261 | 49 | 47 | 0.68 | 13.36 | 13.99 |
| RearAxle | C4 kerb 2g+1g | 411,261 | 139 | 133 | 3.61 | 4.70 | 4.94 |

# FEA iteration: before / after

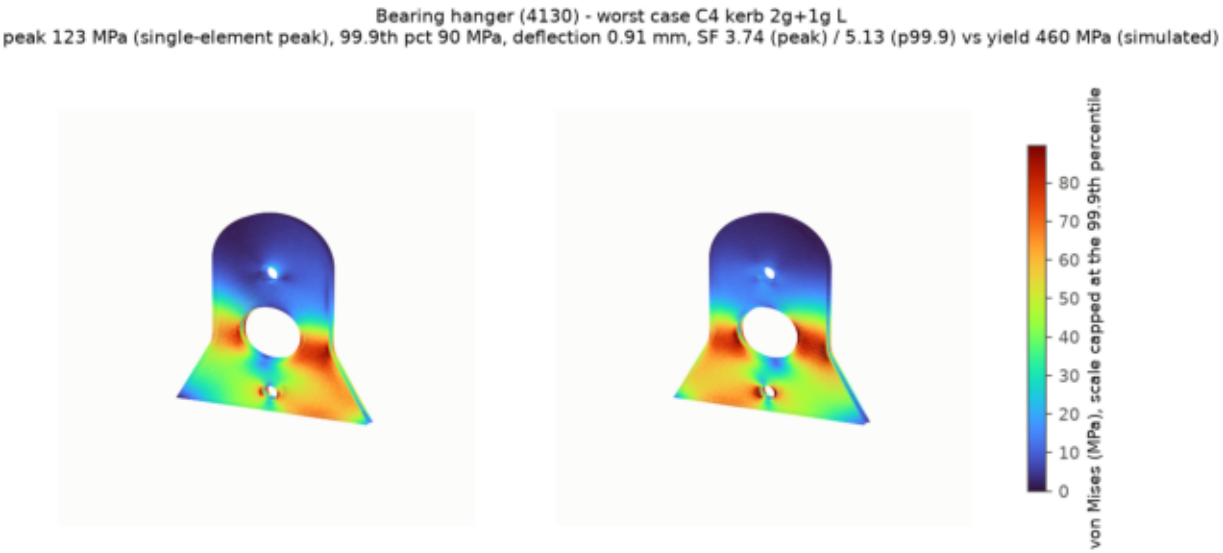




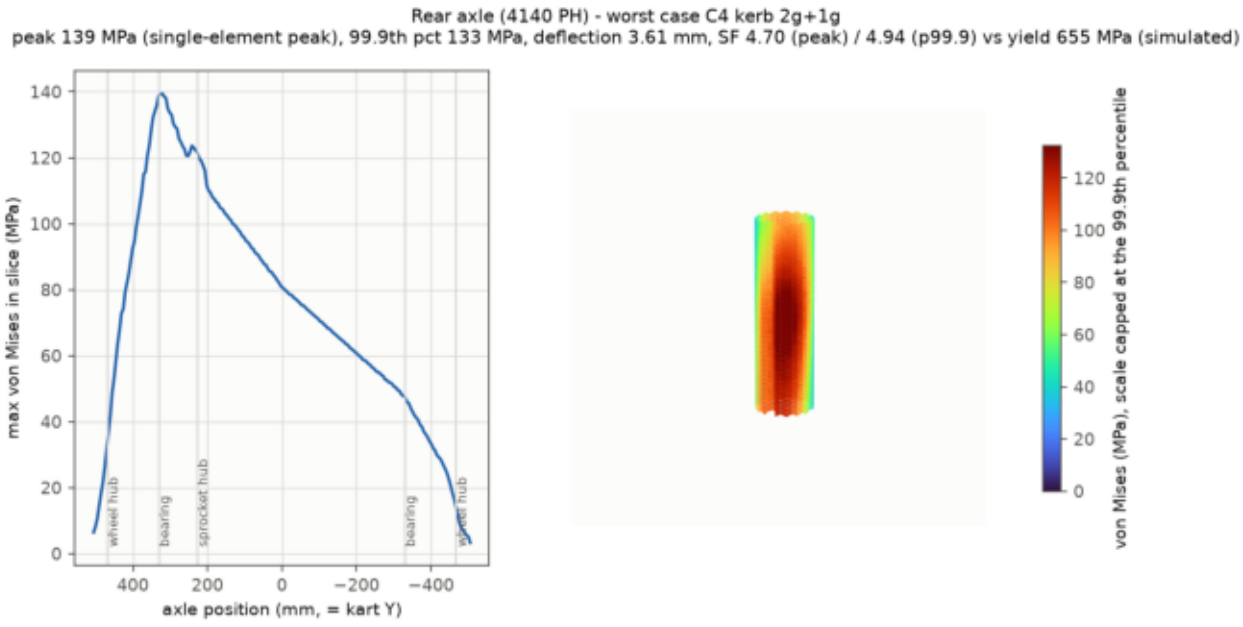

# FEA - Motor-mount bracket (6061-T6)



# FEA - Bearing hanger (4130)



# FEA - Rear axle (4140 PH)



# 6 Assembly checks (Inventor)

Inventor checks: 61 occurrences (26 files), 180 constraints {'Flush': 140, 'Mate': 40}, unhealthy 0, DOF histogram {'0T/0R': 61} (total 0). Parameters {'ChainLine': '250.0 mm', 'ChainAdjust': '0.0 mm'}.

Interference (AnalyzeInterference on all occurrences, > 0.01 mm3): design pose 0 pairs; ChainAdjust -12/-6/+6/+12 mm (chain excluded, since it is modelled tight at 0): 0, 0, 0, 0 pairs; ChainLine 246 / 254 mm (chain included): 0, 0 pairs.

Sprocket alignment: over ChainLine 246-254 mm the solved 13T and 50T mid-planes and the chain plane coincide to 0.0e+00 mm, with axes parallel to 0.0e+00 deg (target < 0.5 mm / 0.1 deg).

The checks found and drove one fix: the reference chassis' rear bumper stays clipped both hanger plates (138 mm3 each) and, at ChainLine 254, the chain; the stays were rerouted.

# 7 Mass and BOM (Inventor)

Inventor mass properties: drivetrain 30.48 kg (CG [157.8, 92.6, 162.0]), rear wheels 5.40 kg. Motor, chain, rims and tyres carry datasheet / catalogue mass overrides.

| item | qty | part no. | description | kg |
|---|---|---|---|---|
| 1 | 2 | GK-101 | Rear-axle bearing hanger, 3/8 in 4130 plate, R25 ble | 2.56 |
| 2 | 1 | GK-102 | Motor-mount base plate 270 x 142 x 6, A36, welded ac | 1.79 |
| 3 | 1 | GK-704 | Caliper bracket 1/4 in A36 plate 68 x 105, TIG-welde | 0.35 |
| 4 | 1 | GK-103 | Motor mount bracket, 6061-T6 machined from solid: 3/ | 1.95 |
| 5 | 1 | GK-301 | Rear live axle 1-1/4 in x 40 in, 4140 pre-hard 28-32 | 6.16 |
| 6 | 1 | GK-201 | Motenergy ME1114 brushless PMSM, 24-72 V, 0.12 Nm/A, | 13.15 |
| 7 | 1 | GK-401 | Sprocket ANSI 40 13T, 7/8 bore, 3/16 key, B-hub 1.56 | 0.19 |
| 8 | 1 | GK-402 | Sprocket ANSI 40 50T, 1-1/4 bore + 1/4 keyway, B-hub | 2.77 |
| 9 | 1 | GK-403 | Roller chain ANSI 40 single strand, 78 links incl. c | 0.60 |
| 10 | 2 | GK-411 | Rear wheel hub 1-1/4 bore x 80 long, 1/4 keyway, 3 h | 1.05 |
| 11 | 2 | GK-421 | Flange housing UCFL206 2-bolt, cast iron, bolt centr | 1.02 |
| 12 | 2 | GK-422 | Insert ball bearing UC206-20, 1-1/4 bore, set-screw | 0.79 |
| 13 | 2 | GK-511 | Square key 1/4 x 1/4 x 25, 1018 CD (axle sprocket hu | 0.02 |
| 14 | 2 | GK-512 | Square key 1/4 x 1/4 x 70, 1018 CD (wheel hubs) | 0.04 |
| 15 | 1 | GK-513 | Square key 3/16 x 3/16 x 19, 1018 CD (motor sprocket | 0.00 |
| 16 | 1 | GK-705 | Chain and sprocket guard, 1.5 mm steel laser cut + b | 1.00 |
| 17 | 2 | GK-412 | Kart rim 5 in dia x 8 in wide, 3 holes on 2.5 in BC, | 2.70 |
| 18 | 2 | GK-413 | Kart slick 11 x 7.10-5 (rear), envelope | 3.40 |
| 19 | 1 | GK-701 | Brake-disc hub 1-1/4 bore, 1/4 keyway, 3 x 1/4 on 2. | 0.26 |
| 20 | 1 | GK-702 | Brake disc 7.1 in (180 mm) x 1/8 in steel, 3-bolt on | 0.56 |
| 21 | 1 | GK-703 | Hydraulic kart brake caliper (single piston, rear) e | 0.45 |
| 22 | 4 | GK-521 | Hex head cap screw 3/8-16 x 1-1/4, SAE J429 grade 8, | 0.11 |
| 23 | 4 | GK-522 | Hex head cap screw 3/8-16 x 1, SAE grade 8 (motor fa | 0.10 |
| 24 | 4 | GK-523 | Hex head cap screw 1/2-13 x 1-1/2, SAE grade 8 (bear | 0.25 |
| 25 | 4 | GK-525 | Nylon-insert locknut 3/8-16, grade 8 | 0.06 |
| 26 | 4 | GK-526 | Nylon-insert locknut 1/2-13, grade 8 | 0.14 |
| 27 | 4 | GK-528 | Flat washer DIN 125 M10 (10.5 x 20 x 2), used under | 0.02 |
|  | 53 | TOTAL (CAD, Inventor) |  | 41.47 |
|  |  | of which designed (welded |  | 13.81 |
|  |  | of which rear rims + tyres |  | 6.10 |
|  |  | NOT MODELLED - electrical |  |  |
|  | 1 | GK-601 | Motor controller, 48 V sine-wave PMSM, sin/cos input | 2.90 |
|  | 1 | GK-602 | 16S LiFePO4 40 Ah pack (51.2 V nominal, 2.05 kWh), w | 20.00 |
|  | 1 | GK-603 | Main contactor 48 V 200 A + 250 A fuse + key switch | 0.60 |
|  | 1 | GK-604 | Hall throttle pedal 0-5 V | 0.45 |
|  | 1 | GK-605 | Phase and battery cable 2 AWG, lugs (estimate) | 1.20 |
|  | 2 | GK-706 | Caliper mounting bolts M8 x 35 class 10.9 + nylon-in |  |
|  | 3 | GK-707 | Disc bolts 1/4-20 x 3/4 grade 8 + nylon-insert nuts |  |
|  | 4 | GK-708 | Guard fixings M6 x 12 button head + washers + rivet |  |
|  | 1 | GK-709 | Brake fluid DOT 4, 0.5 l (for the GK-703 hydraulic k |  |
|  | 4 | GK-710 | Hardened washer ASTM F436 1/2 in, under the UCFL206 |  |

| | | |
|---|---|---|
| | TOTAL incl. electrical sys | 66.62 |

# 8 Assumptions and limitations

Designed and simulated only. No part of this drivetrain has been built or tested; every number is a model prediction.

Vehicle: 185 kg with an 80 kg driver, 60 % rear, CG 280 mm, wheelbase 1040 mm, Crr 0.018, CdA 0.54 m2, mu 1.1 (assumed, typical kart values). Rolling radius 137 mm (11.0 in tyre, 2.7 mm deflection assumed).

Motor data from the Motenergy ME1114 listing (Thunderstruck EV): 24-72 V, 200 A cont. / 420 A peak, 65 Nm peak stall, 0.12 Nm/A, 5000 rpm, 6.5 mOhm, 92 % peak efficiency, 29 lb, 7/8 in shaft with 3/16 key. NOT on the listing, so assumed: rotor inertia 0.012 kg m2, the face bolt pattern (4 x 3/8-16 on 165.1 mm) and the motor's outer shape (envelope model). The DC-equivalent Ke = Kt model ignores field weakening and iron loss (0.3 Nm drag lumped).

Controller (350 A phase / 250 A battery, Sevcon Gen4 Size 4 class) and battery (16S LiFePO4 40 Ah, 30 mOhm) are class assumptions; neither is modelled in CAD (listed separately in the BOM with estimated masses).

Purchased parts without vendor CAD (sprockets, chain, UC206 / UCFL206, hubs, rims, tyres, keys) are modelled from catalogue dimensions (tooth form: ANSI 40 seating curve with straight 20 deg flanks); ratings: chain 13.9 kN min. ultimate and 3.6 kN maximum allowable (Tsubaki RS40), UC206 C 19.5 / C0 11.3 kN. Fasteners and washers are step.parts STEP files (DIN 125 M10 washer used as the 3/8 washer).

Loads are quasi-static limit cases (no dynamic amplification measured). The FEA is linear elastic with nominal geometry, no weld fillets and no keys; the hanger weld zone is checked against an assumed 360 MPa as-welded 4130 allowable. Fatigue is checked analytically for the axle only.

The chassis section is reference geometry. The chain is modelled tight at ChainAdjust = 0. The brake hub, disc, caliper, hubs and rims are catalogue-class envelopes, and the caliper piston / master cylinder sizes are assumed. Seat, steering and the electrical installation are not modelled.